

The role of financial analysts in deterring corporate fraud in China

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Abstract

Existing studies on the determinants of corporate fraud in emerging markets mainly focus on the internal governance mechanism and little is known about the impact of external governance features such as the analyst coverage. Analysts have been recently found to play a direct role in the detection and report of fraud in developed countries such as the US. In this paper, we show that in addition to the role of whistle blower, analysts can effectively enhance the governance quality and deter fraud among the Chinese listed firms. The effect of analyst coverage in reducing the incidence of corporate fraud is more pronounced among non-state-owned enterprises (NSOE), but not among state-owned enterprises (SOE), presumably because NSOEs are more dependent on external equity capital. Additional results also confirm that the regulatory enforcement actions against corporate fraud affect the subsequent performance of SOEs less than NSOEs.

Keywords: Analyst coverage; Corporate fraud; State ownership; China

JEL classification: G32; P37

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1. Introduction

We examine whether financial analyst coverage deters corporate fraud based on the sample of listed firms in China, which is an increasingly influential emerging economy. Corporate fraud can impede China's aspiring economic development since it has serious consequences to stakeholders (e.g. Davidson and Worrell, 1988; Chen et al., 2005), employees (e.g. Zahra et al., 2005), and society (e.g. Szwajkowski, 1985). Existing studies (Yu, 2008; Dyck et al., 2010) confirm that financial analysts contribute to the detection of earnings management and corporate fraud in the US. Effective external monitoring promotes investors confidence and is therefore crucial to the functioning of any capital market. However, the literature on corporate fraud in China has so far focused extensively on the internal governance mechanisms (e.g. Chen et al., 2006; Jia et al., 2009; Ding et al. 2011; Hou and Moore, 2010) and paid less attention to external governance and in particular the role of analysts. As China reforms its capital market to bolster its economic growth, the effectiveness of external monitoring effect such as those provided by financial analyst in curbing corporate fraud warrants investigation. Our study fills this gap in the literature.

Unlike other ex-communist transitional economies, the Chinese government retains control and influence of many listed firms through ownership. Existing studies (e.g. Allen et al., 2005; Chen et al., 2010) suggest that state-owned enterprises (SOEs) in China receive greater financial support (e.g. favorable loans from state banks and government subsidies) from central and/or local governments than non-state-owned enterprises (NSOEs). As a result, NSOEs have greater dependence on external capital from the stock market than SOEs and are therefore also more accountable to external scrutiny. For instance, empirical studies confirm that executive pay is more sensitive to stock return performance (Firth et al., 2006) and cost of equity capital is more sensitive to audit quality (Chen et al., 2011) among NSOEs than SOEs. As a result, we expect corporate fraud impact NSOEs differently

than SOEs.

When corporate fraud is detected, it signals to outside investors that its management lacks credibility and/or that its fundamentals are poorer than expected. As a result, such firms will subsequently have greater difficulty in raising external capital to fund profitable investment opportunities. Due to government financial backing, SOEs in China are less dependent on external funding than NSOEs, and as such, NSOEs will incur greater costs associated with fraud. Financial analysts are sophisticated users of both private and public information, and fraud detection is positively correlated with the number of analysts following a firm. Therefore, financial analyst coverage deters corporate fraud in China more effectively among NSOEs than SOEs.

To test this assertion we conduct empirical analyses on a sample of Chinese listed firms with 8,274 firm-year observations over the period of 2003 to 2008. We acquire enforcement actions against corporate fraud from the China Center for Economic Research (CCER) database. We measure financial analyst coverage by three proxies: the number of analyst following the firm, the number of research reports issued for the firm, and the number of brokerage houses that issue analyst report on the firm. Across all three proxies, we consistently observe a significantly negative relationship between analyst coverage and corporate fraud propensity among NSOEs, but not SOEs. Our findings are also robust to controls for firm characteristics, governance, as well as industry and regional effects. Further analyses reveals NSOEs are associated with more negative firm value decline as well as earnings performance following corporate fraud relative to their SOE counterparts. This confirms that NSOEs firms do incur greater cost upon detection of their fraudulent activity, which reinforces the idea that external monitoring is more effective in deterring fraud against NSOEs.

Our findings have the following implications. First, financial analysts contribute to fraud deterrence even in transitional economies like China where legal enforcement and shareholder

protection are relatively weaker than developed countries like the US. However, the effectiveness of this external corporate governance mechanism depend firms' degree of reliance on external capital. Second, although some studies suggest political connection is beneficial to Chinese listed firms (e.g. Xin and Pearce, 1996; Luo, 2003), we show that it has adverse effect to minority shareholders through the reduction of firms' accountability to external monitoring. Despite some studies suggest that Chinese SOEs are evolving well to adapt to the new market-based economy (e.g. Lin and Germain, 2003; Ralston et al., 2006), such evidence could be confounded by the government financial support available to such firms.

Our paper is organized as follows. Section 2 reviews the literature, explains the institutional background, and develops our hypotheses. Section 3 describes our sample and methodologies. Section 4 presents our empirical findings. Section 5 concludes.

2. Literature and hypotheses

2.1 Corporate fraud and the monitoring effect of analysts

Several antecedents of corporate fraud have been suggested by existing literature. Internal factors includes organization culture (e.g. McKendall and Wanger, 1997), top management (e.g. Baucus, 1994; Ashforth and Anand, 2003), and board composition (e.g. Dunn, 2004). External factors includes environmental hostility (e.g. Baucus and Baucus, 1997), environmental dynamism (e.g. Hansen et al., 1996), industry cultures (e.g. Baucus and Near, 1991), industry concentration (e.g. McKendall and Wanger, 1997), and regulatory condition (e.g. Hou and Moore, 2010). According to Black (2005), corporate fraud can be categorized into two types, i.e. opportunistic and reactive. The former occurs when executives seizes an opportunity for further gain by manipulating disclosure and the latter occurs when executives responds to declining firm performance by window dressing financial statements.

There is mixed evidence on the effect of internal corporate governance mechanism in deterring corporate fraud. Beasley (1996) found reduced financial statement fraud among firms with more outside directors but also observe that the presence of audit committee does not reduce fraud likelihood. Uzun et al (2004) also observes outside directors reduce fraud but the presence of nomination committee increases the propensity of fraud. Denis et al (2006) even observe that executive stock options can tempt managers to commit fraudulent behavior. Indeed, Berenson (2003) suggest that corporate fraud remains uncurbed in the US despite of decades of continued effort to reform corporate governance, to align managers' incentives with shareholders' interest, and impose code of conduct for managerial ethics. Schnatterly (2003) suggest that traditional corporate governance mechanisms such as blockholders, boards, and CEO compensation have limited effect on reducing corporate fraud. Dyck et al (2010) analyze a sample of US corporate fraud cases over the period of 1996 to 2005 reveals that internal governance mechanism, which the literature on fraud deterrence focuses on, accounts for only a third of the fraud detection.

Financial analysts provide a monitoring effect to firms and serves as an external governance mechanism (e.g. Jensen and Meckling, 1976; Healy and Palepu, 2001). They are perceived by managers to be highly influential on share prices (e.g. Graham et al., 2005). Yu (2008) suggest analysts have three distinctive features that make them effective monitors of firms. First, unlike internal governance mechanism designed to protect existing shareholder interest, analysts can provide information that also benefit prospective investors and other market participants. Second, analysts have training in finance and accounting as well as background in industries they cover, which qualifies them to extract information from tedious financial statements and complicated footnotes. Finally, analysts track firms on regular basis and continuously scrutinize financial reporting irregularities, which provide a level of oversight that board members and regulators cannot match. Dyck et al (2010) argue that while analysts do not receive direct monetary compensation for revealing

fraud, they can benefit through enhanced reputation and career prospects (e.g. Hong and Kubik, 2003). They show that financial analysts contribute more to fraud detection than many other categories of external governance mechanisms (such as auditors, client/competitors, shareholders, regulators, SEC) on individual basis.

2.2 Corporate fraud in China

Corporate fraud in China is likely to be motivated by two general factors. First, it may be stimulated by regulatory pressure and financial needs (e.g. Szwajkowski, 1985). For instance, stock exchange listing is only allowed for firms making profit for at least two consecutive years (e.g. Aharony et al., 2000), listing firms are required a minimum of 10% return on equity (ROE) for three continuous years before they can issue additional shares (e.g. Chen and Yuan, 2004), and listed firms making two consecutive years of losses will be placed under special treatment and a further year of loss will see such firms suspended from trading or delisted (e.g. Jiang and Wang, 2008). Thus, these stock exchange listing rules gives listed firms the motivation to carry out managerial misconduct despite of the intention of the Chinese authority to guide equity capital toward well-performing firms. Second, corporate fraud are more likely to exist in dynamic and rapidly evolving environment (e.g. Baucus and Near, 1991). As a transitional economy China has weak legal enforcement and investor protection (e.g. Allen et al., 2005), which enables often encourages firms to exploit and carry out opportunistic behavior.

Fraudulent behavior widespread among Chinese listed firms ranges from providing false statements, delaying information disclosure, to embezzlement (e.g. Chen et al., 2005). The China Securities Regulatory Commission (CSRC) is the primary regulator of securities markets in China and is similar to the Securities and Exchange Commission (SEC) in the US. The CSRC investigates and disciplines corporate fraud as part of the responsibility of CSRC to oversee the securities markets.

The CSRC carries out regular reviews, random inspections, and also responds to information and complaints of fraud allegations from investors, employees, and media. Upon identification of misconduct, the CSRC's enforcement actions could range from internal and public criticism to formal criminal prosecution. The CSRC have been criticized for being ineffective in identifying and prosecuting fraud (e.g. Anderson, 2000) and being influenced by political pressures since it is a ministry-level commission that answers to the state (e.g. Chen et al., 2005; Liebman and Milhaupt, 2008). Similar issue also arise in the US since the SEC is funded by the US Government and is therefore also susceptible to political pressures (e.g. Correia, 2009).

Existing empirical studies of corporate fraud in China reveal some evidence of the influence of corporate governance. For instance, Chen et al. (2006) observe less likelihood of corporate fraud among firms with higher proportion of non-executive directors or with separate person serving as CEO and chairperson. They suggest this implies that independent directors provides more effective monitoring role and that CEO duality encourages abuse of managerial power. Jia et al. (2009) provide further evidence that Chinese firms with larger and more active supervisory board are likely to face more severe CRSC sanctions for fraud. Thus, this strand of literature has focused more on the influence of internal governance mechanism. Hou and Moore (2010) find that residual state ownership in privatized firms holds back the governance quality and increase the occurrence of fraudulent activities. However, as China seeks to modernize its capital market, there is also a need to evaluate effectiveness of external governance mechanism provided by financial analysts in deterring corporate fraudulent behavior.

2.3 State ownership of listed firm in China

Many SOEs in China became partially privatized and issued shares that are traded on the stock exchanges since the early 1990s. However, the government (including central and local) often

retained sufficient shares to maintain control of these listed firms. This approach reflects the socio-political ideology of China and differentiates it from other transitional economies, where the government has relinquished all ownership (e.g., Russia). The SOEs accounts for around two thirds of firms listed on the Shanghai and Shenzhen stock exchanges in mainland China today. Among these firms, the government retains decision rights on crucial issues such as asset disposal, merger and acquisition, and CEO appointment (e.g. Chen et al., 2010). While their NSOE counterparts can focus primarily on shareholder wealth maximization, the Chinese SOE listed firms must also bear the responsibility to carry out the political and social objectives of the government (e.g. Bai et al., 2000). In return, SOEs enjoy greater financial support from the government, which in turn reduces their financial constraint and bankruptcy risk (Faccio et al., 2006).

The Chinese government assists SOE listed firms through channels such as subsidy as well as state-owned bank loan. Subsidies are granted by either central or local government to develop strategic areas such as public utilities and high-tech industries, and to assist financially distressed firms (e.g. Chen et al., 2008). Government officials also have personal incentives to assist SOEs through subsidies to aid the development of local economy, which in turn facilitates their promotion through bureaucratic ranks (e.g. Li, 1998). Chinese state-owned banks often lend SOEs for reasons other than profitability targets, e.g. political, employment, and tax (e.g. Chen et al., 2010). Studies show that lending by these state-owned banks is less prudent than their non-state-owned counterpart (e.g. Jia, 2009) and SOEs as borrowers also have less incentives than NSOEs to repay loans since they are required to provide public services (e.g. Green, 2004). In contrasts, Chinese NSOEs have substantial capital access barrier and loan granting decisions to them are made on more competitive basis (e.g. Linton, 2006).

As a result of government financial support, SOEs are more insulated from the discipline of outside investors through the stock market than their NSOE counterparts. For example, Gul et al.

(2010) find that the low stock price informativeness of listed firms in China relative to other countries, which Morck et al. (2000) document, is more pronounced among SOEs. Chen et al. (2010) document less reported earnings conservatism among Chinese SOEs than their NSOE counterparts. Wang et al. (2008) observe Chinese SOEs are more likely to hire small local auditors, who are assumed to provide lower audit quality. These imply that corporate transparency is especially weak among Chinese SOE listed firms. Conyon and He (2008) show that CEO turnover is less sensitive to share price performance among SOEs in China. This suggests that CEOs of such firms have greater job security and are less accountable to poor stock market performances. Kato and Long (2006a,b) find that Chinese NSOE listed firms are more likely to experience performance improvement following the replacement of CEO. This implies that NSOEs have greater incentives to appoint skillful or talented CEOs that can improve firm performance, possibly because they receive less support from the government.

2.4 Hypothesis development

The aforementioned literature review highlights three key issues related to our study. First, the external monitoring effect of financial analysts contributes to corporate fraud detection. Second, the motivation of Chinese listed firms to commit corporate fraud stems from institutional factors and existing studies in this area focus more on the effect of internal corporate governance mechanism. Third, Chinese SOEs receives more financial assistance from government and are therefore less accountable to outside investors than their NSOE counterparts. We intersect these three issues to examine whether Chinese listed firms' propensity to commit corporate fraud is reduced by financial analyst coverage and whether this effect is moderated by government ownership of listed firms.

There are two reasons why the scrutiny of financial analyst coverage could reduce corporate

fraud. First, analysts facilitate market discipline of firms by providing early warning of value-destroying managerial behaviors to external equity investors, which better enables them push share price downward and raise cost of equity capital on a more timely basis. If this is the case, we expect to observe that analyst coverage is more effective in curbing corporate fraud among firms with greater reliance on external equity capital. Second, this external monitoring serves as a substitute for weaknesses in other corporate governance mechanisms. If this is the case we expect to observe that the negative relationship between analyst coverage and corporate fraud propensity to be more pronounced among poorly governed firms. Corporate governance literature often assume that agency problems are greater in SOEs than NSOEs because of less executive accountability (e.g. Shleifer and Vishny, 1997) and greater state ownership entrenchment effect (Classens et al., 2002) in the former group of firms. Based on this argument, the analyst scrutiny should be more effective among SOEs due to worse corporate governance. However, in the case of China, the difference in government financial support renders the SOE listed firms more accountable to the state and therefore insulates them against the discipline of outside equity investors through the stock market. In contrast, as a result of greater reliance on external capital for funding, the Chinese NSOE listed firms are more concerned about the opinion of external equity investors and thus more sensitive to the monitoring effect. Therefore, we hypothesize that:

H1: *Analyst coverage deters corporate fraud more among NSOEs than SOEs*

Our hypothesis H1 is based on the assumption that financial Chinese NSOE listed firms are more concerned about the opinion of external equity investors to corporate fraud than the SOEs. An interesting research question that stems from this is whether external equity investors respond differently to corporate fraud events of NSOEs relative to the SOEs. Firms that commit corporate fraud signal the market that they have untrustworthy executives and/or have financial distress. This should reduce expected future cash flows or increase discount rate of the firms, which would in turn

lead to decline in share price. Since Chinese SOEs receive greater government funding support and thus enjoys less default risk, we expect share price response to be less negative following fraud event than their NSOE counterparts. As a result of government financial assistance, SOEs are also more likely to recover from distress associated with corporate fraud than the NSOEs. Therefore we hypothesize that:

H2a: *Stock market reactions to fraud announcements are more negative among NSOEs than SOEs.*

H2b: *Earnings performance following fraud is lower among NSOEs than SOEs.*

Existing literature suggest that in an efficient capital market, stock price movement should reflect information about future earnings (e.g. Beaver et al., 1980; Ayers and Freeman, 2000). Thus, if we find empirical evidence that supports both hypotheses H2a and H2b, this would imply that the asymmetrically more negative market reactions following corporate fraud among NSOEs are rational response of investors to the subsequent performance they expect of the firms. Alternatively, we find evidence in support of hypothesis H2a but not H2b, then the investors may have responded to the corporate fraud events irrationally and possibly driven by rumors or herding.

3. Sample and Methodology

3.1 Hypotheses tests

To test hypotheses H1, which predicts a negative relationship between corporate fraud propensity and financial analyst coverage, we apply logistic regression analyses based on Equation 1 below:

$$\begin{aligned}
\text{Fraud}_{i,t} = & \alpha_0 + \alpha_1 \text{Coverage}_{i,t-1} + \alpha_2 \text{NSOE}_{i,t-1} + \alpha_3 \text{Coverage}_{i,t-1} \times \text{NSOE}_{i,t-1} \\
& + \sum_{k=1}^k \alpha_{k+3} \text{Control}_k + \varepsilon
\end{aligned} \tag{1}$$

The dependent variable *Fraud* equals 1 if enforcement actions against corporate fraud occurred in the current fiscal year and 0 otherwise. All independent variables are lagged for 1-year to deal with the causality issue. *Coverage* is separately proxied by the number of financial analysts following the firm, the number of analyst reports issued for the firm, and the number of brokerage house that issue analyst reports on the firms. *NSOE* equals 1 if the firm is non-state-owned enterprise and 0 otherwise. *Coverage* and *NSOE* are thereby interacted. *Size* is firm size measured as log market capitalization. A set of control variables are also incorporated. *Growth* is firm growth measured as price-to-book ratio. *ST* equals 1 for firms on the verge of special treatment (i.e. those with two consecutive years of losses) and 0 otherwise. *Foreign* equals 1 if the proportion of institutional foreign shares is above 0% and 0 otherwise. *FundOwn* equals 1 if the proportion of shares held by mutual fund is above yearly cross-sectional median or 0 otherwise. *OwnCon* is ownership concentration measured by the Herfindahl index and based on the ownership held by the ten largest shareholders in the firm. *Duality* equals 1 for firms with CEO also serving as board chairman and 0 otherwise. *Bmeet* equals 1 for firms with above yearly cross-sectional median number of board meetings and 0 otherwise. *Bsize* equals to 1 for firms with board size above the yearly cross-sectional median and 0 otherwise. *Bindep* equals 1 for firms whose proportion of independent directors is above the yearly cross-sectional median and 0 otherwise. *Sbsize* equals 1 for firms with supervisory board size above the yearly cross-sectional median and 0 otherwise. Industry and region dummies are also included to control for the impact of industry and regional effects. The industry dummies are constructed based on the first two digits of the GICS(Global Industry Classification Standard) codes. The region dummies are constructed by following Frith et al. (2006), in which the firms are grouped into four different regions by the levels of economic development: 1. Shanghai and Shenzhen; 2. The more developed areas including the open cities and provinces along the coast; 3. The inland provinces; and 4. the least developed area in the north-western part of the country. The coefficient α_1 indicates the relationship between fraud

propensity and analyst coverage among SOEs. The coefficient α_3 indicates the incremental relationship between fraud propensity and analyst coverage among NSOEs. If $\alpha_3 < 0$ and is statistically significant, this indicates the analyst coverage reduces corporate fraud significantly more among NSOEs, which confirms our prediction in H1.

To test hypotheses H2a and H2b, which predict that share price response to fraud events are more negative and earnings performance following fraud are lower among NSOEs, we implement the following regression analyses in the corporate fraud sub-sample:

$$CAR_{i,t} = \beta_0 + \beta_1 NSOE + \sum_{k=1}^k \beta_{k+1} Control_k + \varepsilon \quad (2)$$

$$EPS_{i,t} = \beta_0 + \beta_1 NSOE + \sum_{k=1}^k \beta_{k+1} Control_k + \varepsilon \quad (3)$$

The dependent variable of Equation 2 is the cumulative abnormal return (CAR) around the announcement day of the enforcement actions against corporate fraud. We calculate abnormal returns as firm-specific return less the market index returns either on day 0 or over a five day window (-2 to $+2$) of the event. All other variables are defined as in Equation 1. Coefficient $\beta_1 < 0$ indicates that NSOEs are associated with more negative stock returns upon disclosure of corporate fraud than their SOE counterparts, which confirms our hypothesis H2a. The dependent variable of Equation 3 is the earnings per share (EPS) in the fiscal year following enforcement actions against corporate fraud. We apply either raw or industry adjusted EPS. Coefficient $\gamma_1 < 0$ indicates that NSOEs are associated with lower earnings performance in the year after the corporate fraud than their SOE counterparts, which confirms our hypothesis H2b.

3.2 Sample description, summary statistics, and correlation analyses

We obtain archival data required for our analyses from either CCER (China Centre for Economic Research) or CSMAR (China Stock Market and Accounting Research). These include regulatory enforcement actions against corporate fraud being disclosed, firms' ownership status, firms' characteristics and performance, as well as firms' corporate governance variables. We identified 409 fraud cases over the six year sample period of 2003 to 2008 where valid data for all other variables required in our analysis are available. These include firm size, market-to-book ratio, special treatment status, foreign ownership, mutual fund ownership, ownership concentration, CEO duality, number of board meetings, board size, board independence, and supervisory board size.

Table 1 presents the summary statistics of the variables used in our empirical analyses. Corporate fraud cases (*Fraud*) accounts for 4.5% of our total number of firm-year observations. Within our sample the average number of analyst reports issued to a firm (*Report*) is 4.481 per year, the average number of analysts following a firm (*Analysts*) is 2.872 per year, and the average number of brokerage houses that issue analyst reports on a firm (*Broker*) is 3.766 per year. Non-state-owned enterprise (*NSOE*) accounts for nearly a third of our total observation. The average firm size based on market capitalization (*Size*) is over 845 million RMB and on the average this is three times above the book value of the firms as indicated by market-to-book ratio (*Growth*). Around 10% of our total observations are classified as special treatment (*ST*), i.e. firms reporting two consecutive years of loss. While 7% of firms in our sample have foreign ownership, only less than 3% have mutual fund ownership.

[insert Table 1 here]

Table 2 presents the correlation analyses of the variables used in our study. *Fraud* has significantly negative correlation with the three analyst coverage variables *Report*, *Analyst*, and *Broker*. It is also significantly higher among NSOE listed firms, possibly because such firms have less government financial support and thus more often resort to fraud perhaps to fulfil the stock exchange listing rules. Firms that are smaller, have higher growth, or suffer loss tend to have more information asymmetry and

therefore commits fraud more frequently. Mutual fund ownership is significantly correlated with less fraud cases, which implies that institutional investors exert a monitoring effect to firms. Ownership concentration also reduces corporate fraud possibly because large block shareholders have more influence to discipline firms. Finally, firms with board of directors that meets more frequently have less fraud, which is consistent with the board imposing internal governance function.

[insert Table 2 here]

4. Empirical findings

4.1 Test of hypothesis H1

Table 3 presents the logistic regression analyses of the relationship between corporate fraud propensity and financial analyst coverage, which is proxied by the number of analysts following the firm. In the SOE sub-sample, the coefficient for *Coverage* is negative but not statistically significant (-0.025 , $t\text{-stat} = -0.91$). In the NSOE sub-sample, however, the coefficient for *Coverage* is significantly negative (-0.131 , $t\text{-stat} = -2.60$). These observations suggest that on a split sample basis, higher analyst coverage is associated with lower corporate fraud propensity only in the NSOE group, which supports our prediction in hypothesis H1. Since the SOE sub-sample has nearly twice the number of observations as the NSOE sub-sample, the observation of an analyst coverage effect being significant only in the latter sub-sample cannot be attributed to any sample size bias. If anything, the smaller sample size of the NSOE sub-sample biases us against finding results in support of our hypothesis H1. In full-sample analysis, the coefficient for $Coverage \times NSOE$ is significantly negative (-0.098 , $t\text{-stat} = -1.86$), which indicates that fraud propensity is incrementally lower among NSOEs with higher analyst coverage. These results consistently indicates that the negative relationship between corporate fraud propensity and financial analyst coverage exist mainly in

Chinese NSOE listed firms, which also confirms the prediction in our hypothesis H1.

[insert Table 3 here]

Our findings are robust to the control of various firm characteristics and corporate governance variables, as well as industry and regional effect. Throughout Table 3, we observe consistent and significant relationship between corporate fraud propensity and *Size*, *ST*, *OwnCon*, and *Bmeet*. The negative coefficient for *Size* and the positive coefficient for *ST* suggest that smaller and loss firms are more likely to commit corporate fraud, possibly because they have more information asymmetry or are in financial distress. The negative coefficient for *OwnCon* and *Bmeet* indicates that firms with more concentrated ownership and more frequent board of directors meeting are less likely to commit fraud. This suggests that internal governance mechanisms provided by block holders and board of directors has some effect in curbing fraud in both SOE and NSOEs. Thus, Table 3 essentially provides an interesting contrast in the effectiveness between external and internal governance mechanisms for Chinese listed firms. External governance is effective only for NSOEs and internal governance is effective in both SOEs and NSOEs.

Tables 4 and 5 provides further robustness checks to our analyses in Table 3 by proxying analyst coverage as the number of reports issued for the firm and the number of brokerage houses that issues analyst reports on the firm respectively. A single analyst can issue more than one research reports for the firm they follow. This usually occurs when there is arrival of new information and the analysts wish to revise their opinion. Thus, the number of reports essentially captures that activeness of the analyst in monitoring the firm. The analysts that follow firms can be employed by two types of brokerage houses, i.e. those that underwrite the securities of the firm and those that do not. Analysts of the latter group are more independent and less influenced by the conflict of interest to market the securities of the firm they follow. Thus, the number of brokerage house captures the level of

independent analyst coverage of the firm. Consistent with Table 3, we observe in Tables 4 and 5 that the coefficient for *Coverage* is significant only in the NSOE (-0.091 , $t\text{-stat} = -2.40$ and -0.110 , $t\text{-stat} = -3.57$ respectively) sub-sample but not the SOE counterpart (0.00 , $t\text{-stat} = 0.00$ and -0.033 , $t\text{-stat} = -1.53$). The coefficients for the interactive term *Coverage* $\times$ *NSOE* are significantly negative in both Tables 4 and 5, which is consistent with the findings of Table 3. These findings confirm that our evidence in support of hypothesis H1 is robust not only to various control variables but also to alternative proxies of financial analyst coverage.

[insert Table 4 here]

[insert Table 5 here]

4.2 Test of hypotheses H2a and H2b

Table 6 presents our results from the test of hypothesis H2a, which predicts that the stock market reacts more negatively to the corporate fraud committed by NSOEs than SOEs. In Panel A we observe that the SOE listed firms are associated with 0.7% decline in value relative to the market index on the day that enforcement actions against their fraudulent behavior are disclosed. On the other hand, the NSOEs are associated with 1.7% decline in value relative to the market portfolio upon announcement of enforcement actions. The difference between the two is statistically significant and similar pattern is observed using a longer five (-2 , $+2$) day event window period. Panel B shows that based on the day 0 and the days -2 to $+2$ window abnormal returns, the coefficient pertaining to *NSOE* are -0.012 ($t\text{-stat} = -2.56$) and -0.022 ($t\text{-stat} = -1.79$) respectively. This further confirms that the market reaction to corporate fraud is more negative among NSOEs than their SOE counterparts after controlling for various firm characteristics and corporate governance variables. On the whole, Table 6 confirms our prediction in hypothesis H2a that NSOEs incur more negative economic consequences

following corporate fraud. This is consistent with external investors attaching a higher discount rate to NSOEs following corporate fraud possibly because the lack of government financial support makes it more difficult for them to recover.

[insert Table 6]

Table 7 reports our test of hypothesis H2b, which predicts that the earnings performance in the year following corporate fraud would be lower among NSOEs than SOEs. Indeed, both Panels A and B reveals that NSOEs raw and industry median adjusted earnings per share are significantly lower than their SOE counterparts one year after the corporate fraud event, which confirms our prediction in hypothesis H2b. The economic significance in the regression evidence is such that EPS is approximately 0.2 lower among NSOEs, and the findings are very similar for raw and industry median adjusted EPS. Mistrust and loss of reputation affects fraud committing firms' ability to acquire external capital, thereby affecting their ability to fund operations and investment, which in turn reduces their earnings performance. With less government financial support, this negative impact is likely to be greater among the NSOEs. The observation in Table 7 of significantly lower earnings performance among NSOEs subsequent to fraud also suggests that the more negative market reaction to such firms upon enforcement actions against fraud is a rationale anticipation of poorer future cash flows.

[insert Table 7 here]

4.3 Discussion

The empirical data provided strong support for our central hypotheses. We acknowledge

alternative interpretations of our findings, and as such which we discuss such alternative explanations in this subsection. First, fraud is more common among NSOEs (Table 2). The smaller subsample of fraudulent cases among SOEs may reduce statistical significance of analyst coverage. We checked this possibility by matching the number of cases of fraud by SOEs and NSOEs with propensity score techniques, and this matched sample provided results that are not materially different from what we have reported above.

Second, there may be differences in state prosecution, settlement and non-reporting of fraud among SOEs and NSOEs, possibly related to bribes or other protection afforded to SOEs. If so, these confounding determinants of reported fraud that are not observed among SOEs may mitigate the statistical significance of reported fraud for SOEs relative to that for NSOEs. We are not able to test this possibility directly, but the evidence provided sheds light on the likelihood of such an interpretation. Most notably, other factors that affect the prevalence of reported fraud, such as size, special treatment and ownership concentration (defined in the appendix) show very similar effects for SOEs and NSOEs (see, e.g., Table 3). This evidence suggests that there are not material unobserved events for SOEs versus NSOEs. By contrast, the effect of analyst coverage is massively different for SOEs versus NSOEs.

Third, as discussed in section 2 above, there are different types of fraud that fall within opportunistic (such as misreporting) and reactive (such as window dressing) categories (Black, 2005). We do not have a massive sample of fraud cases and sufficient details with each case to analyze the different types fraud separately. Nevertheless, we do not have any theoretical reason to expect different types of fraud would be related to analyst coverage in ways that would undo the findings in our empirical tests. Further research could examine this issue with different or extended datasets.

Fourth, there are different ways to define the variables used in our analysis, such as what

constitutes being classified as being a state-owned firm by different ownership levels or key state shareholders. Moreover, there are differences in the amount of time that certain firms have been non-state-owned and the amount of time the fraud went unreported. Our findings pertaining to analyst coverage for SOEs and NSOEs are robust across a very wide range of specifications. Further research could examine other aspects of corporate governance across different variable definitions and the duration of ownership, among other things.

5. Conclusion

We provide empirical evidence that in addition to the whistle blower role already documented in the literature, financial analyst in China also play an effective external monitoring role to deter corporate fraud among NSOE listed firms but not the SOEs listed firms. The evidence is consistent with NSOEs being more sensitive to the discipline of external equity investors, possibly to the lack of government financial support. Further analyses reveal that NSOEs are indeed associated with more negative market reactions after corporate fraud. The data are therefore consistent with the dependence of external capital rendering the mistrust and loss of reputation due to fraud more costly to NSOEs. Our findings are robust to controls of firm characteristics and corporate governance variables, industry and regional effects as well as different specification of analyst coverage.

Existing literature suggest that political connection and government support can be beneficial to Chinese listed firms in terms of market benefit (e.g. Davies et al., 1995), competitive advantages (e.g. Tsang, 1998), and firm performance (e.g. Nee, 1992; Peng and Luo, 2000). Xin and Pearce (1996) argue that political connection is a substitute for insufficient institutional infrastructure. Atuahene-Gima and Li (2002) argue that it facilitates business in an uncertain environment. Lou (2003) suggests that political connections provide flexible resource allocation in a factor mobility

constrained environment. In our case, we confirm that government support reduces the negative economic consequences experienced by SOEs in the context of corporate fraud. However, it also reduces the effectiveness of external scrutiny exerted by external governance mechanisms such as financial analysts. The reduction in exposure to stock market influence we observe among SOEs is broadly consistent to the finding of Kato and Long (2006a,b) and Conyon and He (2008) that the inverse relationship between CEO turnover and firm performance is less pronounced among state controlled Chinese listed firms. Our finding does not reject the value of managerial political connection in a transitional economy, but reveals an additional costs to minority shareholders it incurs.

The efficient allocation of financial resources is a key determinant of economic development and growth. Therefore, as China aspires to become one of the largest economies in the world, its capital market is set to play a vital role. The experiences in developed countries over the past decade, from corporate scandals to financial crisis, have revealed the importance of corporate stewardship and governance to the well-being of the capital market and therefore the wider economy. Therefore, developing and strengthening external governance mechanisms through sources such as financial analysts would be essential to improve outside investors' confidence, which in turn enhances financial market liquidity. Academic studies that identify existing and potential problems could offer useful input to practitioners and regulators to formulate future directions suitable for China's capital market development. Our study provides insight from this perspective.

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Table 1: Summary statistics

This table presents the summary statistics of the variables used in our analysis. Our sample period covers 2003 to 2008. *Fraud* equals 1 if enforcement actions against corporate fraud occurred in the current fiscal year and 0 otherwise. *Report* is the number of analyst reports issued for the firm. *Analyst* is the number of financial analysts following the firm. *Broker* is the number of brokerage house that issue analyst reports on the firms. *NSOE* equals 1 if the firm is non-state-owned enterprise and 0 otherwise. *Size* is firm size measured as log market capitalization. *Growth* is firm growth measured as price-to-book ratio. *ST* equals 1 for firms on the verge of special treatment (i.e. those with two consecutive years of losses) and 0 otherwise. *Foreign* equals 1 if the proportion of institutional foreign shares is above 0% and 0 otherwise. *FundOwn* equals 1 if the proportion of shares held by mutual fund is above yearly cross-sectional median or 0 otherwise. *OwnCon* is ownership concentration measured by the Herfindahl index and based on the ownership held by the ten largest shareholders in the firm. *Dualisty* equals 1 for firms with CEO also serving as board chairman and 0 otherwise. *Bmeet* equals 1 for firms with above median number of board meetings and 0 otherwise. *Bsize* equals to 1 for firms with board size above the cross-sectional median and 0 otherwise. *Bindep* equals 1 for firms whose proportion of independent directors is above the cross-sectional median and 0 otherwise. *SBsize* equals 1 for firms with supervisory board size above the cross-sectional median and 0 otherwise.

Panel A

Variables	Mean	Std.dev.	Median	Obs.
<i>Fraud</i>	0.045	0.208	0.000	8274
<i>Report</i>	4.481	10.447	0.000	8274
<i>Analyst</i>	2.872	5.585	0.000	8274
<i>Broker</i>	3.766	6.111	1.000	8274
<i>NSOE</i>	0.334	0.472	0.000	8274
<i>Size</i>	20.556	1.082	20.408	8274
<i>Growth</i>	3.451	3.971	2.344	8274
<i>ST</i>	0.092	0.289	0.000	8274
<i>Foreign</i>	0.070	0.255	0.000	8274
<i>FundOwn</i>	0.026	0.060	0.000	8274
<i>OwnCon</i>	0.196	0.131	0.161	8274
<i>Duality</i>	0.010	0.099	0.000	8274
<i>Bmeet</i>	0.573	0.495	1.000	8274
<i>Bsize</i>	0.372	0.483	0.000	8274
<i>Bindep</i>	0.883	0.321	1.000	8274
<i>SBsize</i>	0.888	0.315	1.000	8274

Panel B

	SOE				NSOE			
	Mean	Std.dev.	Median	Obs.	Mean	Std.dev.	Median	Obs.
<i>Fraud</i>	0.0358	0.1857	0	5510	0.0644	0.2455	0	2764
<i>Report</i>	4.7432	10.9093	1	5510	3.9573	9.4375	0	2764
<i>Analyst</i>	3.0276	5.7993	1	5510	2.5630	5.1188	0	2764
<i>Broker</i>	3.9374	6.3398	1	5510	3.4237	5.6130	1	2764
<i>Size</i>	20.6717	1.1055	20.5195	5510	20.3248	0.9948	20.1925	2764
<i>Growth</i>	3.2006	3.4233	2.2457	5510	3.9512	4.8438	2.5960	2764
<i>ST</i>	0.0693	0.2540	0	5510	0.1378	0.3448	0	2764
<i>Foreign</i>	0.0543	0.2266	0	5510	0.1017	0.3023	0	2764
<i>FundOwn</i>	0.0273	0.0616	0.0004	5510	0.0229	0.0550	0	2764
<i>OwnCon</i>	0.2210	0.1377	0.1952	5510	0.1462	0.0993	0.1203	2764
<i>Duality</i>	0.0069	0.0828	0	5510	0.0159	0.1252	0	2764
<i>Bmeet</i>	0.5931	0.4913	1	5510	0.5315	0.4991	1	2764
<i>Bsize</i>	0.4149	0.4927	0	5510	0.2865	0.4522	0	2764
<i>Bindep</i>	0.8652	0.3416	1	5510	0.9200	0.2713	1	2764
<i>SBsize</i>	0.9169	0.2761	1	5510	0.8318	0.3741	1	2764

Table 2: Correlation analysis

This table presents the correlation analysis of the variables used in our analysis. Our sample period covers 2003 to 2008. *Fraud* equals 1 if enforcement actions against corporate fraud occurred in the current fiscal year and 0 otherwise. *Report* is the number of analyst reports issued for the firm. *Analyst* is the number of financial analysts following the firm. *Broker* is the number of brokerage house that issue analyst reports on the firms. *NSOE* equals 1 if the firm is non-state-owned enterprise and 0 otherwise. *Size* is firm size measured as log market capitalization. *Growth* is firm growth measured as price-to-book ratio. *ST* equals 1 for firms on the verge of special treatment (i.e. those with two consecutive years of losses) and 0 otherwise. *Foreign* equals 1 if the proportion of shares held by foreign shareholders is above yearly cross-sectional median and 0 otherwise. *FundOwn* equals 1 if the proportion of shares held by mutual fund is above yearly cross-sectional median or 0 otherwise. *OwnCon* is ownership concentration measured by the Herfindahl index and based on the ownership held by the ten largest shareholders in the firm. *Dualisty* equals 1 for firms with CEO also serving as board chairman and 0 otherwise. *Bmeet* equals 1 for firms with above median number of board meetings and 0 otherwise. *Bsize* equals to 1 for firms with board size above the cross-sectional median and 0 otherwise. *Bindep* equals 1 for firms whose proportion of independent directors is above the cross-sectional median and 0 otherwise. *SBSize* equals 1 for firms with supervisory board size above the cross-sectional median and 0 otherwise.* indicates statistical significance at 5% level.

	<i>Fraud</i>	<i>Report</i>	<i>Analyst</i>	<i>Broker</i>	<i>NSOE</i>	<i>Size</i>	<i>Growth</i>	<i>ST</i>	<i>Foreign</i>	<i>FundOwn</i>	<i>OwnCon</i>	<i>Duality</i>	<i>Bmeet</i>	<i>Bsize</i>	<i>Bindep</i>	<i>SBSize</i>
<i>Fraud</i>	1															
<i>Report</i>	-0.0640*	1														
<i>Analyst</i>	-0.0769*	0.9678*	1													
<i>Broker</i>	-0.0889*	0.7419*	0.9406*	1												
<i>NSOE</i>	0.0612*	-0.0086	-0.0094	0.0012	1											
<i>Size</i>	-0.1071*	0.5149*	0.5458*	0.5764*	-0.1420*	1										
<i>Growth</i>	0.0264*	0.0146	0.0163	0.0213	0.0705*	0.1596*	1									
<i>ST</i>	0.1229*	-0.1150*	-0.1340*	-0.1570*	0.1124*	-0.2313*	0.1660*	1								
<i>Foreign</i>	-0.0127	0.0380*	0.0513*	0.0665*	0.0877*	0.0183	0.0008	-0.0206	1							
<i>FundOwn</i>	-0.0671*	0.5721*	0.5944*	0.6282*	-0.0065	0.4773*	0.1107*	-0.1125*	0.0108	1						
<i>FundCon</i>	-0.0707*	0.0260*	0.0310*	0.0468*	-0.2724*	0.005	-0.0470*	-0.1097*	-0.0482*	-0.0569*	1					
<i>Duality</i>	-0.0039	0.0282*	0.0168	0.0056	0.0386*	0.0087	0.0154	0.0288*	0.0267	0.0079	-0.0431*	1				
<i>Bmeet</i>	-0.0574*	-0.0384*	-0.0335*	-0.0248*	-0.0669*	-0.0528*	-0.0375*	-0.0533*	0.0186	-0.0295*	0.0452*	-0.0165	1			
<i>Bsize</i>	-0.0101	-0.0097	-0.0104	-0.0289*	-0.1174*	0.0660*	-0.0393*	-0.0520*	0.0510*	0.0056	0.0293*	0.0086	0.0501*	1		
<i>Bindep</i>	-0.019	0.0722*	0.0872*	0.1113*	0.1330*	0.0339*	-0.0595*	0.0018	0.0293*	0.1298*	-0.0815*	-0.0048	-0.0397*	-0.0011	1	
<i>SBSize</i>	-0.018	0.0134	0.0066	-0.0059	-0.1072*	0.0771*	-0.0079	-0.0444*	-0.0315*	-0.0006	0.0284*	-0.0262*	-0.0223	0.0515*	-0.003	1

Table 3: The number of analysts and fraud

This table presents the logistic regression analysis of the relationship between corporate fraud propensity and financial analyst coverage as follows.

$$Fraud_{i,t} = \alpha_0 + \alpha_1 Analyst_{i,t-1} + \alpha_2 NSOE_{i,t-1} + \alpha_3 Analyst_{i,t-1} \times NSOE_{i,t-1} + \sum_{k=1}^K \alpha_{k+3} Control_k + \varepsilon$$

The dependent variable *Fraud* equals 1 if enforcement actions against corporate fraud occurred in the current fiscal year and 0 otherwise. *Analyst* is the number of financial analysts following the firm. *NSOE* equals 1 if the firm is non-state-owned enterprise and 0 otherwise. Other control variables are defined in the Appendix. The sample period covers 2003 to 2008. Our t-statistics are adjusted for heteroskedasticity. *, **, *** indicate 10%, 5%, and 1% level significance respectively.

	SOE sub-sample		NSOE sub-sample		Full sample	
	Coeff	t-stat	Coeff	t-stat	Coeff	t-stat
<i>Analyst</i>	-0.025	(-0.91)	-0.131 ***	(-2.60)	-0.027	(-1.03)
<i>NSOE</i>					0.257 ***	(3.16)
<i>Analyst</i> × <i>NSOE</i>					-0.098 *	(-1.86)
<i>Size</i>	-0.214 ***	(-4.05)	-0.203 ***	(-3.32)	-0.198 ***	(-4.74)
<i>Growth</i>	0.023 **	(2.10)	-0.003	(-0.44)	0.008	(1.19)
<i>ST</i>	0.410 ***	(3.40)	0.258 **	(2.34)	0.361 ***	(4.44)
<i>Foreign</i>	0.132	(0.59)	-0.143	(-0.75)	-0.015	(-0.09)
<i>FundOwn</i>	-2.146	(-1.48)	-1.611	(-0.55)	-1.936	(-1.37)
<i>OwnCon</i>	-0.905 ***	(-2.68)	-2.109 ***	(-3.51)	-1.232 ***	(-4.18)
<i>Duality</i>	-0.119	(-0.29)	-0.061	(-0.22)	-0.076	(-0.33)
<i>Bmeet</i>	-0.262 ***	(-3.57)	-0.299 ***	(-3.51)	-0.277 ***	(-4.86)
<i>Bsize</i>	0.104	(1.14)	-0.102	(-1.04)	0.017	(0.25)
<i>Bindep</i>	-0.081	(-0.76)	-0.197	(-1.57)	-0.130	(-1.57)
<i>SBSIZE</i>	-0.121	(-0.96)	-0.072	(-0.65)	-0.098	(-1.13)
<i>Intercept</i>	3.033 **	(2.51)	4.191 ***	(3.41)	3.575 ***	(4.08)
Industry effect	Yes		Yes		Yes	
Region effect	Yes		Yes		Yes	
Pseudo R2	0.120		0.151		0.135	
Obs.	5510		2764		8274	

Table 4: The number of analysts reports and fraud

This table presents the logistic regression analysis of the relationship between corporate fraud propensity and financial analyst coverage as follows:

$$Fraud_{i,t} = \alpha_0 + \alpha_1 Report_{i,t-1} + \alpha_2 NSOE_{i,t-1} + \alpha_3 Report_{i,t-1} \times NSOE_{i,t-1} + \sum_{k=1}^K \alpha_{k+3} Control_k + \varepsilon$$

The dependent variable *Fraud* equals 1 if enforcement actions against corporate fraud occurred in the current fiscal year and 0 otherwise. *Report* is the number of analyst reports issued for the firm. *NSOE* equals 1 if the firm is non-state-owned enterprise and 0 otherwise. Other control variables are defined in the Appendix. The sample period covers 2003 to 2008. Our t-statistics are adjusted for heteroskedasticity. *, **, *** indicate 10%, 5%, and 1% level significance respectively.

	SOE sub-sample		NSOE sub-sample		Full sample	
	Coeff	t-stat	Coeff	t-stat	Coeff	t-stat
<i>Report</i>	0.000	(0.00)	-0.091**	(-2.40)	-0.003	(-0.22)
<i>NSOE</i>					0.264***	(3.30)
<i>Report</i> × <i>NSOE</i>					-0.079***	(-2.17)
<i>Size</i>	-0.233***	(-4.35)	-0.203**	(-3.32)	-0.209***	(-4.96)
<i>Growth</i>	0.024**	(2.17)	-0.003	(-0.41)	0.009	(1.24)
<i>ST</i>	0.410***	(3.40)	0.267**	(2.41)	0.364***	(4.48)
<i>Foreign</i>	0.119	(0.53)	-0.147	(-0.78)	-0.023	(-0.14)
<i>FundOwn</i>	-3.487**	(-2.02)	-1.614	(-0.56)	-2.841*	(-1.81)
<i>OwnCon</i>	-0.943***	(-2.79)	-2.125***	(-3.55)	-1.267***	(-4.29)
<i>Duality</i>	-0.136	(-0.33)	-0.053	(-0.19)	-0.077	(-0.33)
<i>Bmeet</i>	-0.258***	(-3.52)	-0.304***	(-3.58)	-0.276***	(-4.85)
<i>Bsize</i>	0.100	(1.10)	-0.109	(-1.12)	0.010	(0.15)
<i>Bindep</i>	-0.086	(-0.81)	-0.200	(-1.59)	-0.136	(-1.63)
<i>SBSIZE</i>	-0.117	(-0.92)	-0.063	(-0.57)	-0.093	(-1.08)
<i>Intercept</i>	2.952***	(2.60)	3.990***	(3.04)	3.033***	(3.26)
Industry effect	Yes		Yes		Yes	
Region effect	Yes		Yes		Yes	
Pseudo R2	0.118		0.150		0.134	
Obs.	5510		2764		8274	

Table 5: The number of brokerage houses and fraud

This table presents the logistic regression analysis of the relationship between corporate fraud propensity and financial analyst coverage as follows:

$$Fraud_{i,t} = \alpha_0 + \alpha_1 Broker_{i,t-1} + \alpha_2 NSOE_{i,t-1} + \alpha_3 Broker_{i,t-1} \times NSOE_{i,t-1} + \sum_{k=1}^k \alpha_{k+3} Control_k + \varepsilon$$

The dependent variable *Fraud* equals 1 if enforcement actions against corporate fraud occurred in the current fiscal year and 0 otherwise. *Broker* is the number of brokerage that issue analyst reports on the firms. *NSOE* equals 1 if the firm is non-state-owned enterprise and 0 otherwise. Other control variables are defined in the Appendix. The sample period covers 2003 to 2008. Our t-statistics are adjusted for heteroskedasticity. *, **, *** indicate 10%, 5%, and 1% level significance respectively.

	SOE sub-sample		NSOE sub-sample		Full sample	
	Coeff	t-stat	Coeff	t-stat	Coeff	t-stat
<i>Broker</i>	-0.033	(-1.53)	-0.110***	(-3.57)	-0.036*	(-1.70)
<i>NSOE</i>					0.269***	(3.27)
<i>Broker</i> × <i>NSOE</i>					-0.068**	(-1.99)
Size	-0.201***	(-3.75)	-0.191***	(-3.11)	-0.186***	(-4.41)
Growth	0.023**	(2.06)	-0.004	(-0.49)	0.008	(1.13)
ST	0.403***	(3.34)	0.236**	(2.12)	0.347***	(4.26)
Foreign	0.138	(0.61)	-0.116	(-0.60)	0.001	(0.01)
FundOwn	-1.429	(-1.06)	-0.896	(-0.33)	-1.186	(-0.91)
OwnCon	-0.874***	(-2.60)	-2.037***	(-3.41)	-1.189***	(-4.06)
Duality	-0.116	(-0.28)	-0.071	(-0.26)	-0.081	(-0.35)
Bmeet	-0.261***	(-3.55)	-0.291***	(-3.41)	-0.273***	(-4.76)
Bsize	0.108	(1.17)	-0.102	(-1.04)	0.019	(0.27)
Bindep	-0.074	(-0.68)	-0.189	(-1.51)	-0.122	(-1.47)
SBSIZE	-0.131	(-1.02)	-0.099	(-0.87)	-0.116	(-1.33)
Intercept	2.768**	(2.27)	4.065***	(3.33)	2.935***	(3.19)
Industry effect	Yes		Yes		Yes	
Region effect	Yes		Yes		Yes	
Pseudo R2	0.122		0.155		0.138	
Obs.	5510		2764		8274	

Table 6: Market reaction to fraud announcements

This table presents tests of stock market reaction to corporate fraud announcements. We apply the sub-sample of firms with corporate fraud. The dependent variable is cumulative abnormal return calculated as firm-specific return less market index returns. Panel A applies day 0 return and Panel B applies cumulative returns over 5-day window (−2, +2). *NSOE* equals 1 if the firm is non-state-owned enterprise and 0 otherwise. *SOE* stands for state-owned enterprises. Other variables are defined in the appendix. The sample period covers 2003 to 2008. The t-statistics are adjusted for heteroskedasticity. *, **, *** indicate 10%, 5%, and 1% level significance respectively.

Panel A				
	CAR[0]	Obs	CAR[-2,+2]	Obs
Combined sub-sample	-0.011	396	-0.017	338
SOE sub-sample	-0.007	238	-0.010	209
NSOE sub-sample	-0.017	158	-0.029	129
Difference	0.009 ***		0.019 ***	
t-stat	2.854		2.363	
Panel B				
	CAR[0]		CAR[-2,+2]	
	coeff	t-stat	coeff	t-stat
<i>NSOE</i>	-0.012 **	(-2.56)	-0.022 *	(-1.79)
<i>Size</i>	0.000	(0.17)	0.016 **	(2.59)
<i>Growth</i>	0.000	(0.94)	0.000	(-0.25)
<i>ST</i>	0.005	(1.08)	0.018	(1.39)
<i>Foreign</i>	0.009	(0.76)	0.025	(1.45)
<i>FundOwn</i>	-0.092	(-1.35)	-0.285 *	(-1.85)
<i>OwnCon</i>	-0.023	(-1.22)	-0.050	(-1.02)
<i>Duality</i>	0.001	(0.08)	-0.025	(-0.96)
<i>Bmeet</i>	0.001	(0.27)	0.030 ***	(2.77)
<i>Bsize</i>	0.001	(0.15)	-0.015	(-1.09)
<i>Bindep</i>	0.002	(0.38)	-0.017	(-1.41)
<i>SBSIZE</i>	-0.003	(-0.44)	0.016	(0.85)
<i>Intercept</i>	-0.018	(-0.31)	-0.335 ***	(-2.68)
Industry clustering	Yes		Yes	
Adj R ²	-0.002		0.058	
Obs	267		225	

Table 7: Earnings performance following corporate fraud

This table presents analyses of earnings performance in the year following corporate fraud. We apply the sub-sample of firms with corporate fraud. Panel A reports comparison tests. Panel B reports regression results. We test both raw earnings per share (EPS) and industry median adjusted earnings per share (IndAdj EPS) in Panel B. IndAdj EPS is the difference between a firm's raw EPS and the median value of EPS of all firms from the same industry for the same year. *NSOE* equals 1 if the firm is non-state-owned enterprise and 0 otherwise. SOE stands for state-owned enterprises. Other variables are defined in the appendix. The sample period covers 2003 to 2008. The t-statistics are adjusted for heteroskedasticity. *, **, *** indicate 10%, 5%, and 1% level significance respectively.

Panel A				
	EPS	Obs	IndAdj EPS	Obs
Combined sub-sample	-0.116	381	-0.288	381
SOE sub-sample	0.075	199	-0.092	199
NSOE sub-sample	-0.326	182	-0.501	182
Difference	0.401 ***		0.409 ***	
t-stat	4.167		4.327	
Panel B				
	EPS coeff	t-stat	IndAdj EPS coeff	tstat
<i>NSOE</i>	-0.199 ***	(-2.96)	-0.209 ***	(-3.17)
<i>Size</i>	0.081 **	(2.42)	0.067 ***	(2.12)
<i>Growth</i>	-0.009	(-1.50)	-0.010 *	(-1.66)
<i>ST</i>	0.137 *	(1.89)	0.123 *	(1.77)
<i>Foreign</i>	-0.015	(-0.13)	0.019	(0.18)
<i>FundOwn</i>	4.128 ***	(2.97)	4.231 ***	(3.14)
<i>OwnCon</i>	0.295	(1.51)	0.274	(1.45)
<i>Duality</i>	0.145 *	(1.74)	0.132	(1.61)
<i>Bmeet</i>	0.004	(0.06)	0.002	(0.02)
<i>Bsize</i>	0.002	(0.02)	0.012	(0.16)
<i>Bindep</i>	-0.013	(-0.17)	-0.028	(-0.37)
<i>SBSIZE</i>	-0.276 ***	(-2.99)	-0.251 ***	(-2.82)
<i>Intercept</i>	-1.439 **	(-2.05)	-1.322 **	(-1.99)
Industry clustering	Yes		Yes	
Adj R ²	0.116		0.114	
Obs	381		381	

Appendix: Variable Definitions

<i>Fraud</i>	A dummy variable assigned to 1 if the firm is subject to a regulatory enforcement against fraud, and 0 otherwise
<i>Report</i>	The number of analyst reports issued for the firm.
<i>Analyst</i>	The number of financial analysts following the firm.
<i>Broker</i>	The of brokerage house that issue analyst reports on the firms.
<i>NSOE</i>	A dummy variable equals 1 if the firm is non-state-owned enterprise and 0 otherwise.
<i>Size</i>	Firm size measured as log market capitalization.
<i>Growth</i>	Firm growth measured as price-to-book ratio.
<i>ST</i>	A dummy variable equals 1 for firms on the verge of special treatment (i.e. those with two consecutive years of losses) and 0 otherwise
<i>Foreign</i>	A dummy variable equals 1 if the proportion of institutional foreign shares is above 0% and 0 otherwise.
<i>FundOwn</i>	A dummy variable equals 1 if the proportion of shares held by mutual fund is above yearly cross-sectional median or 0 otherwise.
<i>OwnCon</i>	Ownership concentration measured by the Herfindahl index and based on the ownership held by the ten largest shareholders in the firm.
<i>Duality</i>	A dummy variable equals 1 for firms with CEO also serving as board chairman and 0 otherwise.
<i>Bmeet</i>	A dummy variable equals 1 for firms with above median number of board meetings and 0 otherwise.
<i>Bsize</i>	A dummy variable equals to 1 for firms with board size above the cross-sectional median and 0 otherwise.
<i>Bindep</i>	A dummy variable equals 1 for firms whose proportion of independent directors is above the cross-sectional median and 0 otherwise.

The following industry and region dummies are also incorporated in our empirical analyses:

The industry dummies are constructed based on the first two digits of the GICS (Global Industry Classification Standard) codes. The region dummies are constructed by following Firth et al. (2006), in which the firms are grouped into four different regions by the levels of economic development: 1. Shanghai and Shenzhen; 2. The more developed areas including the open cities and provinces along the coast; 3. The inland provinces; and 4. The least developed area in the north-western part of the country.